From Interpretable to Black-Box: Dynamic Model Identification for Small USV Control Design

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***Abstract*—Designing the low-level speed and heading loops of a small uncrewed surface vehicle (USV) requires a dynamic model, and it is rarely clear how much model the job needs. For vessels under seven meters with a single thruster and rudder, tow-tank testing is impractical and parameters must be identified from limited field trials. We treat sufficiency as a property of a stated design job, taking that job to be the proportional-integral plus feedforward stabilization loops of a production autopilot, and identify three model levels against that target: first-order linear SISO transfer functions, a nonlinear state-space form with quadratic drag, and a NARX multilayer perceptron. First-order models reach $R^2 = 0.93$ in surge and $R^2 = 0.83$ in yaw rate, enough to fix a PI+FF compensator. A margin-based test then reports where that model stops being good enough: a speed controller designed from it behaves as intended only between 0.35 and 1.28 m/s, since below that band the model error exceeds the stability margin and above it the loop stays stable but loses more than half its intended bandwidth. The NARX model simulates the comparison trial at $R^2 = 0.47$ in speed and 0.85 in yaw rate once the logs are curated, settling at 1.30 m/s where the vessel does 1.22, but it plateaus near 1.3 m/s beyond the speeds its corpus covers. The structured levels do not share that limit, because their equilibrium follows from the equation rather than from samples.**

***Index Terms*—uncrewed surface vehicle, system identification, model spectrum, control-oriented modeling, robustness margin, underactuated vessels**

I. Introduction

Small uncrewed surface vehicles, here vessels under seven meters driven by a single thruster and rudder, are increasingly deployed for survey, inspection and autonomy research. Every such mission rests on a working low-level controller: a speed loop on the throttle and a yaw-rate loop on the rudder, each holding a setpoint while a guidance layer above does the navigating. In production autopilots such as ArduPilot these are proportional-integral loops with an added feedforward term. Designing, simulating and tuning that pair of loops is the task this paper addresses, and the task against which every modeling decision below is judged.

The identification literature offers everything from two-parameter response models to maneuvering polynomials carrying dozens of hydrodynamic coefficients, but no guidance on which this job requires. Fixing the compensator structure makes the question answerable, because it turns concrete: does a change of model level move the gains, the achievable bandwidth or the tuning procedure of that loop? A model that alters none of the three is inert however much better it fits. Taking the most detailed model available is not a safe default either, since added terms degrade the conditioning of the fit [1], [2] and each one needs excitation to identify, which converts fidelity directly into field trials on a platform where trials are the binding constraint.

The nearest prior attempt to weigh that trade shows where the difficulty sits. Sonnenburg and Woolsey [3] evaluate a cost function on held-out maneuvers, tabulate it per model and per regime, then select by judgment, discarding the best-scoring model because two extra parameters are not worth the slight edge. Their cost function scores the benefit of added parameters and no comparable metric exists for the cost. Skelton put the criterion in control terms in 1989, that model errors need not be small but only appropriate for the control design [4], and it has stood since without becoming computable. Making it computable for the loops above is the contribution here.

Our scope is the single thruster plus rudder vessel with field-identified parameters. Underactuated rudder steering is the interesting case, because control authority is an explicit function of forward speed, so the steering and speed channels cannot be treated as independent design problems. We identify three model levels from common field data, compare them with cross-model diagnostics, and give a margin-based test reporting the speed range over which added fidelity stops changing the loop design. The deliverable is a procedure rather than a model. Prior studies for this vehicle class compare candidate structures and then select one, after which the rest stop contributing; we keep the set. The models buy loop structure, verification in simulation, relative comparison of variants and a rehearsed tuning procedure. They do not buy gains that transfer without field tuning, and we say so plainly.

II. Closely Related Work

The minimal control-oriented description of a steered vessel is Nomoto's first-order steering model [5], still the standard starting point in the modern formulation [6], with the identification tradition founded by Åström and Källström [7]. Physics-based maneuvering models such as Abkowitz's [8] include coefficients with physical meaning but resist identification from field data: over the maneuvers a small vessel can realistically perform the regressors are nearly collinear, so many coefficient sets fit about equally well and none can be trusted on its own however well the model predicts as a whole [1], [2], [9]. For small craft the closest prior work is Sonnenburg and Woolsey [3], [10], with related efforts spanning autonomous surface craft and small workboats [11]–[15].

Data-driven and grey-box marine models span recurrent architectures [16], physics-informed networks [17] and gradient-boosted ensembles [18], and our Level 3 follows the neural identification literature [19] and its delayed-input variants [20]. Evaluation for control design has to be by simulation, in which the model is initialized once and thereafter runs on its own output, since that is what a design loop asks of it. Two gaps follow: sufficiency is resolved by selecting one model rather than retaining the set, and the small rudder-steered vessel is underrepresented relative to differential-thrust research platforms.

III. Model Spectrum Methodology

Each level is judged by what it contributes to the PI-plus-feedforward loop of Sec. I rather than by goodness of fit: Level 1 settles the loop structure, Level 2 says where nonlinear feedforward is needed and Level 3 bounds what a black-box alternative would buy.

Level 1 treats surge and yaw as decoupled first-order channels, $G_v(s) = K_v/(\tau_v s + 1)$ and $G_r(s) = K_r/(\tau_r s + 1)$, mapping normalized throttle to speed and normalized rudder to yaw rate. Both are identified as discrete autoregressive models with exogenous input, ARX(n_a, n_b) with $n_a = n_b = 1$, so that $y(k) = a_1 y(k-1) + b_1 u(k-1)$, fitted by least squares on the 10 Hz record and converted to continuous time. τ is the open-loop pole the controller works against, so it sets the scale of closed-loop bandwidth worth asking for, and K sets the loop gain. A PI structure follows, with internal model control tuning fixing its gains as $k_p = \tau/(K\lambda)$ and $k_i = 1/(K\lambda)$ for a chosen closed-loop time constant λ.

Level 2 keeps the decoupled structure but replaces linear damping in surge with quadratic drag,

$$\dot{v} = k_T\, u - d_2\, v|v|, \tag{1}$$

$$\dot{r} = k_\delta\, \delta - d_r\, r. \tag{2}$$

Identification is two-phase. Fitting (1) jointly for k_T and d_2 is poorly conditioned, because thrust and drag are large and nearly cancelling during steady runs, so the data constrains their difference far better than either term. Instead d_2 is estimated from coasting segments where $u = 0$ and (1) reduces to $\dot{v} = -d_2 v|v|$, then k_T follows from the powered segments. The yaw channel keeps a constant k_δ, a local approximation rather than a physical claim: rudder force grows with the square of flow speed, so the value identified holds near the speed at which the rudder steps were flown.

Level 3 is a fixed-tap residual NARX multilayer perceptron with 19 lagged inputs, one hidden layer of 64 tanh units and two outputs which are state increments rather than absolute states. State history extends 0.5 s into the past, throttle history 2 s and rudder history 1 s, a project-specific choice consistent with delayed-input NARX models [20]. Being multi-input multi-output, it is free to discover the surge-yaw coupling absent in Levels 1 and 2. Its role is to bound the benefit of added complexity at the available data scale.

Every fit reported here is by simulation: the model is initialized once from measured history and thereafter runs on its own output, replaying the throttle and rudder actually recorded, to the end of the segment without correction. Level 3 initializes from 2 s of measured history, the span of its longest input tap. No controller is present, so the question asked is whether the model can stand in for the vessel. Fit is reported per channel as $R^2 = 1 - \sum_k (y_k - \hat{y}_k)^2 / \sum_k (y_k - \bar{y})^2$, so zero is the score of predicting the channel mean and negative values are worse than that.

A. Cross-Model Diagnostics

The value of the spectrum is the comparison across levels rather than the three fits individually. Scoring one structure on held-out data is a weaker check here, because a maneuvering model can pass a prediction test while its coefficients remain physically meaningless [1], and a coefficient that cannot be interpreted cannot inform a control gain. The four diagnostics are not an arbitrary list: Levels 1 and 2 differ in exactly one term, the form of the surge damping, and share exactly two simplifying assumptions, that the channels are decoupled and that rudder authority does not vary with speed. One diagnostic tests the term that separates the levels, two test the shared assumptions, and the fourth asks whether any of it reaches the compensator.

Diagnostic 1, speed-dependent rudder gain. Replace the constant k_δ in (2) with a speed-scaled $k_\delta v$ and test whether the yaw-rate fit improves. Gain scheduling is well within a production autopilot's capability, so the question is whether the model can say what schedule to use; a constant k_δ cannot. The alternative tested is partial, since Nomoto scaling gives $k_\delta \propto v^2$ with $d_r \propto v$ [6] while the regression scales only the input gain and only linearly.

Diagnostic 2, drag nonlinearity. The form of the surge damping is the only structural difference between Levels 1 and 2, so this is the warrant for Level 2 existing at all. Fit linear and quadratic damping to the coasting deceleration and compare; a dominant quadratic term calls for nonlinear feedforward on the speed loop.

Diagnostic 3, rudder-induced surge drag. The reverse coupling to Diagnostic 1, so together they test decoupling in both directions. Correlate δ^2 against the surge residual of the Level-2 fit. A correlation would make every rudder command a disturbance to the speed loop.

Diagnostic 4, is the added fidelity inside the robustness margin? The first three ask whether a term is real; this asks whether a real term changes the design. The robustness margin of a compensator is a budget for model error, so the test is whether the Level-1 to Level-2 discrepancy fits inside the margin of a controller designed from Level 1 alone. Design a nominal internal model control PI from Level 1 only, $C(s) = (\tau_v s+1)/(K_v \lambda s)$, giving a nominal loop $L = 1/(\lambda s)$ and complementary sensitivity $T(s) = 1/(\lambda s+1)$. Linearizing

Fig. 1. The experimental platform, a Pro Boat Blackjack 42 monohull of 1.07 m length. Thrust comes from a single propeller and steering from a single rudder, both at the stern.

Level 2 about $v_0 > 0$, where $\partial(v|v|)/\partial v = 2v_0$, gives a speed-scheduled first-order plant

$$G_2(s, v_0) = \frac{K_2(v_0)}{\tau_2(v_0)s + 1}, \quad \tau_2 = \frac{1}{2d_2 v_0}, \quad K_2 = \frac{k_T}{2d_2 v_0}, \tag{3}$$

in which gain and time constant both fall as $1/v_0$. Form the multiplicative error $\ell(\omega, v_0) = |G_2(j\omega, v_0)/G_1(j\omega) - 1|$ and test robust stability, $\|\ell T\|_\infty < 1$. That test alone proves insufficient: quadratic drag is stabilizing, so at high speed the Level-1 design becomes sluggish rather than unstable and the stability test keeps passing. We therefore add a performance criterion, comparing the achieved closed-loop bandwidth of the Level-1 compensator on $G_2(s, v_0)$ against the design intent $1/\lambda$. The pair bounds sufficiency from both sides.

IV. Experimental Platform and Data

The platform (Fig. 1) is a Pro Boat Blackjack 42, a production radio-controlled monohull of 1.07 m length instrumented to approximately 5 kg, single-screw with one rudder. Guidance, control and logging run on a Cube Orange executing ArduRover 4.6.3, with ground speed from a HERE3 GNSS receiver. The parameter set confirms the actuation: `SERVO1` is assigned to ground steering and `SERVO3` to throttle, every other output disabled, so there is no differential-thrust path. Commands are logged at 1100–1900 μs PWM and normalized to $[-1, 1]$ about the 1500 μs neutral point; outputs are GPS speed at 5 Hz and IMU yaw rate at 137 Hz. Trials are flown in manual mode, so logged commands are open-loop operator inputs. The autopilot's stabilization layer is the design target: two loops, `ATC_SPEED_*` and `ATC_STR_RAT_*`, each nominally a PID with feedforward but with derivative gain set to zero on this vehicle, so the deployed compensator is PI plus feedforward. The configured cruise speed is 5 m/s, so the identification trial covers roughly the lower half of the intended speed range.

Multi-rate streams are aligned to a common 10 Hz grid by linear interpolation and segmented at mode changes and quality discontinuities. Levels 1 and 2 are identified from a single step-response trial: a 130 s window from a 341 s recording, throttle 0–39%, speed 0–2.9 m/s, 1,300 samples. Level 3 draws on a larger corpus of 44,133 samples in 70 slow-region segments from 65 source logs, where the slow-region rule excludes a segment once speed stays above 3.4028 m/s for at least 1 s. Stationary records dominated the raw logs and were removed on a physical-response criterion, that the vessel demonstrably responds to its controls, which is what makes the Level-3 fit usable at all. Allocation is grouped by source log. The 28 logs held back for evaluation are not a clean held-out set: they combine partitions used for early stopping, for model-family selection, and a nominal test set that was opened before the checkpoint was fixed, so those metrics are optimistic and we treat them as descriptive. The frozen model is additionally evaluated on the same 130 s trial as Levels 1 and 2. That trial is not part of the Level-3 corpus, confirmed by cross-correlating it against every corpus segment, so it is unseen by Level 3 while Levels 1 and 2 are identified from it. Its 20-sample lag history leaves 1,279 scored rows against 1,300.

V. Results

Least squares on the step-response trial gives $G_v(s) = 6.42/(1.95s + 1)$ and $G_r(s) = 0.771/(0.471s + 1)$, with simulation $R^2 = 0.930$ in surge and 0.826 in yaw rate. These are in-sample figures: the trial is short enough that withholding part of it would leave too little excitation to identify against, so fit and score both use all 1,300 samples, and the same holds for Level 2. The surge and yaw time constants differ by a factor of four, which alone settles the loop structure.

Two-phase identification gives $\dot{v} = 3.963\,u - 0.438\,v|v|$ and $\dot{r} = 1.474\,\delta - 1.912\,r$, with $R^2 = 0.909$ in surge and 0.820 in yaw rate. Both sit marginally below Level 1 on this trial, which is expected, since Level 1 minimizes exactly this error on exactly this data while Level 2 takes its drag coefficient from coasting alone.

A. Level 3

Trained on the curated logs, Level 3 simulates the common trial at $R^2 = 0.47$ in speed and 0.85 in yaw rate. Fitted instead to the raw corpus it is unusable, returning -1.99 in speed, and the reason is visible in steady state rather than in the trajectory. A long simulation is governed by where the model settles: hold throttle constant and whatever speed the model converges to is what the run reports. Levels 1 and 2 have an equilibrium by construction, $v = K_v u$ and $v = \sqrt{k_T u/d_2}$, and both track the measured steady speeds with root-mean-square errors of 0.33 and 0.18 m/s. The uncurated network settles below zero at a throttle of 0.20, where the vessel measurably does 1.22 m/s, and its steady-state map is not monotone in throttle, which no physical vessel can be. Curation moves that settling speed to 1.30 m/s.

What remains after curation is an envelope limit rather than a defect, and Fig. 2 shows its shape. The curated network plateaus near 1.3 m/s for any throttle above 0.15, returning 1.24 m/s at a throttle of 0.39 where the vessel does 2.09, because the corpus is capped at 3.14 m/s and holds almost no sustained fast running. A network can only locate the speed

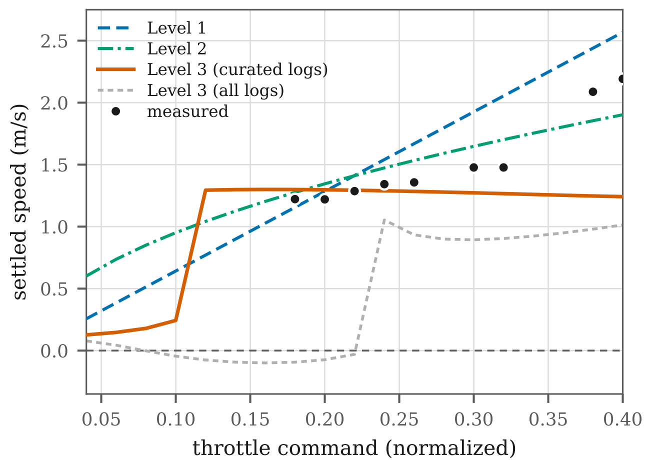

Fig. 2. Steady-state response of each level, obtained by driving the model from rest at constant throttle for 60 s and recording where it settles, against speeds measured on the trial. Levels 1 and 2 have an equilibrium by construction and track the measurements. The curated network tracks them inside the corpus envelope and plateaus beyond it.

at which acceleration vanishes if the data visited it. The structured levels do not share the limit: their equilibrium follows from the form of the equation, so it is defined at operating points no trial ever reached. Within the envelope the corpus covers, the curated network is a serviceable simulator; outside it, the structured models are the only ones that extrapolate.

B. Cross-Model Diagnostic Summary

Diagnostic 1 finds no improvement from a speed-dependent rudder gain ($\Delta R^2 = -0.03$). The trial holds forward speed nearly constant during the rudder steps, so δv and δ are very nearly the same regressor, with a correlation of 0.99, and no fit can separate them; the partial scaling noted above is a second candidate explanation. Either way the result flags a gap in data coverage rather than absence of the effect, and rudder sweeps at separated fixed speeds are what would settle it.

Diagnostic 2 finds quadratic damping dominant, with $\Delta R^2 = +0.20$ over the linear fit, $d_1 = -0.015$ and $d_2 = 0.448\ \mathrm{m}^{-1}$ on the coasting segments. The two-phase identification returns $d_2 = 0.438\ \mathrm{m}^{-1}$ from the same segments, a weak consistency check rather than an independent one.

Diagnostic 3 finds rudder-induced surge drag negligible, with a correlation of $r = -0.08$ between δ^2 and the surge residual. The coupling term is not warranted.

Diagnostic 4 is reported in Fig. 3, computed with $\lambda = 1.0$ s, and returns an interval rather than a threshold. Robust stability fails below $v_0 = 0.35$ m/s, a bound with the closed form $v_{\mathrm{lo}} = k_T/(4d_2K_v)$, independent of λ because $|T| \leq 1$ places the supremum of $\ell|T|$ at DC. The mechanism is the $1/v_0$ singularity in (3): quadratic drag provides no linear damping at zero speed. At the upper end the stability test never fails, and at 2.9 m/s the model error is still within budget at $\|\ell T\|_\infty = 0.76$. What fails is performance: the Level-1 compensator on the Level-2 plant retains only 20% of its intended bandwidth at

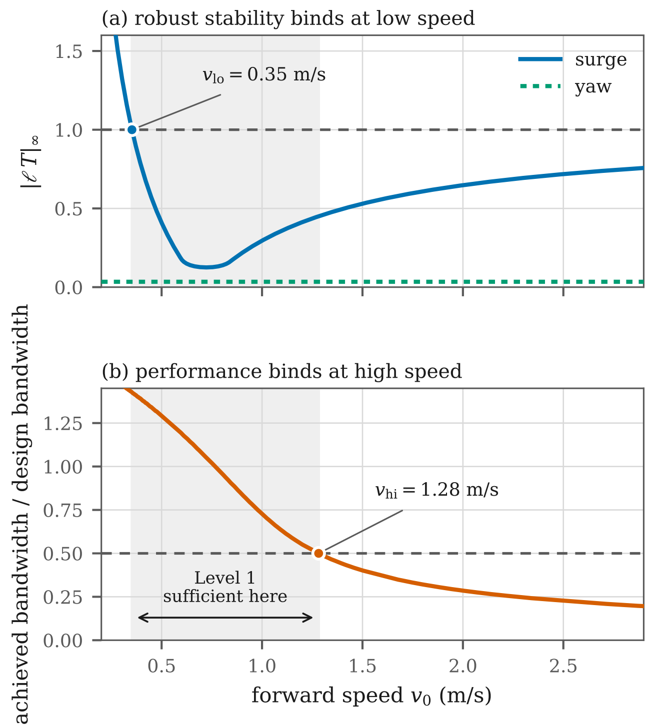

Fig. 3. Diagnostic 4. (a) Robust stability of the Level-1 compensator against the Level-2 linearization binds at low speed, where the quadratic-drag linearization is singular. (b) Achieved closed-loop bandwidth binds at high speed. The shaded band is the interval over which the Level-1 model supports the design. The yaw channel (a, dotted) never approaches the limit.

2.9 m/s, crossing 50% at $v_{\mathrm{hi}} = 1.28$ m/s. The two bounds are not equally firm. The lower follows from $\|\ell T\|_\infty = 1$; the upper depends on choosing 50% of design bandwidth as the point where degradation becomes unacceptable, which is a convention rather than a result, since a 70% criterion moves it to 1.03 m/s and a 30% criterion to 1.91 m/s. Both are insensitive to λ: sweeping it from 0.5 to 3 s moves v_{hi} only between 1.23 and 1.61 m/s. For yaw, Level 2 linearizes to $0.771/(0.523s + 1)$ against Level 1's $0.771/(0.471s + 1)$, identical DC gain and an 11% difference in time constant, giving $\|\ell T\|_\infty = 0.034$. Level 2 buys nothing for yaw control across the whole envelope.

VI. Discussion

For this vessel at the speeds tested, surge and yaw are weakly coupled, so decoupled SISO controllers are a defensible starting point. Quadratic drag is significant, so a speed-hold controller needs nonlinear feedforward and a purely linear one will show speed-dependent steady-state error. Whether the rudder gain scales with speed can be neither confirmed nor denied from the identification trial. No single model exposes all of this: Level 1 gives no reason to suspect the drag is quadratic, Level 2 does not reveal that its extra structure is inert for yaw, and Level 3 alone would have carried the stationary records into the model unremarked.

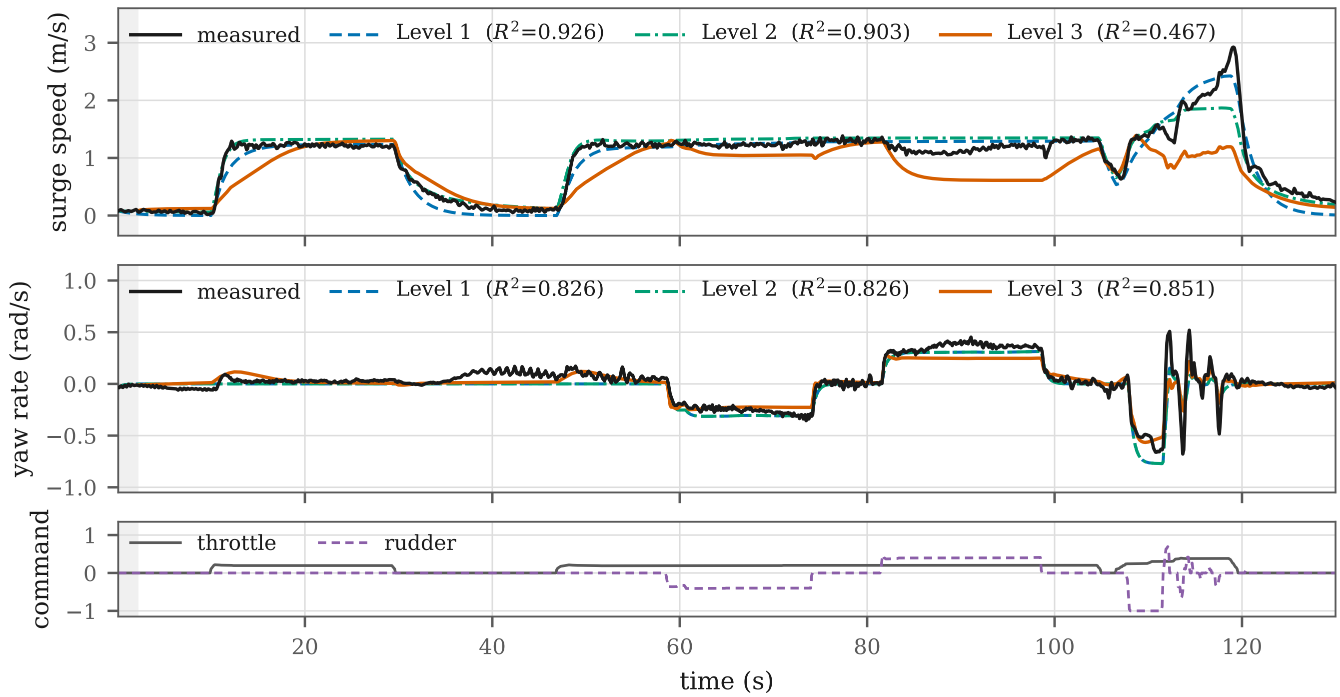

Fig. 4. All three levels simulated on the common 130 s trial, with Level 3 trained on the curated logs. Level 3 begins after 2 s of measured lag initialization, so it scores 1,279 rows against 1,300 for Levels 1 and 2, and all three are scored on those same rows.

Level 1 supports a baseline PI with directly identified gains, valid over the interval Diagnostic 4 returns. Level 2 justifies nonlinear feedforward on the surge channel and says at what speed it becomes necessary; on yaw it changes nothing, so the added structure should not be carried into the design. Level 3 bounds what lies beyond the fixed architecture, and its curated performance says the data now supports a black-box simulator inside the slow region but not beyond it. None of the three delivers gains that work on the water without adjustment. What the spectrum buys is the loop structure, a simulation to verify it in, a basis for comparing variants and a tuning session that can be planned rather than improvised.

It is tempting to read Fig. 4 as showing that physical structure beats data volume, since Level 3 saw 28,023 training samples against the 1,300 used for Levels 1 and 2. That reading is wrong twice over. The levels were fitted and scored under different protocols, so the comparison cannot separate structure from sampling; and refitting Levels 1 and 2 on Level 3's own training partition and scoring all three on the same held-out segments reverses the ordering, giving R^2 of $+0.53$, $+0.64$ and $+0.84$ respectively. On equal terms the network is the better predictor. The defensible claim is about coverage rather than model class: a structured model does not need the experiment to contain steady running, because setting $\dot{v} = 0$ in (1) returns $v = \sqrt{k_T u/d_2}$ whether or not any trial ever sat there. The same theme runs through Diagnostic 1, where the trial could not separate δ from δv, and through Diagnostic 4, where sufficiency is a property of the operating point. In each case what a model can tell a designer is bounded by what the experiment made visible.

Limitations. Levels 1 and 2 rest on a single trial covering 0–2.9 m/s and throttle to 39%, so the identified parameters are not a characterization across the envelope, and Diagnostic 4 inherits that limitation. The common-trial comparison rests on one 130 s record, enough to demonstrate the behavior but not to quantify how often it occurs. Level 3's corpus is restricted to the slow region by construction, so nothing here speaks to behavior near the planing transition, and with a configured cruise speed of 5 m/s no level is characterized over the upper half of the intended range. All results come from a single vessel.

VII. Conclusion

We have presented a three-level model spectrum for small underactuated USVs, identified from field trials and posed throughout against one concrete question: what does a practitioner need in order to design, simulate and tune the speed and yaw-rate loops of a production autopilot? The spectrum collectively answers that question better than any single model does. The margin-based test converts Skelton's qualitative criterion into a computed interval, returning $[0.35, 1.28]$ m/s for the surge channel and reporting that the same test is satisfied everywhere on the yaw channel, where the added Level-2 structure is inert. The black-box level, once its logs are curated, is a serviceable simulator inside the region its data covers and silent outside it, which is the practical sense in which the simpler levels remain necessary.

Future work runs in three directions. The first is data: dedicated rudder sweeps at separated fixed speeds, which is what breaks the collinearity Diagnostic 1 could not overcome, and sustained running above the slow-region cap, which is what would extend the black-box envelope. The second is a stronger basis for cross-level comparison, since the present common trial is a single record and a multi-trial evaluation set would let the comparison be quantified rather than demonstrated; the sensitivity of each level to training-set size belongs here too. The third is to treat the robustness margin as a formal sufficiency criterion across the spectrum, computing for each design decision the level at which further fidelity is provably inert, replacing expert judgment with a computable bound on how much model a given autopilot loop requires.

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